Cache-Line Aligned MemTable Layout and Deterministic SSTable Flush Compaction in a Rust LSM-Tree Implementation

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Abstract—Log-Structured Merge-Trees (LSM-trees) achieve high write throughput by buffering updates in an in-memory MemTable and periodically flushing sorted runs to disk-resident SSTables. In modern cloud workloads, contention on cache lines and the cost of atomic synchronization dominate latency during MemTable insertion and SSTable compaction. This paper presents a Rust implementation that (i) aligns the MemTable's B-Tree nodes to cache-line boundaries, (ii) employs lock-free ring buffers for write ordering, and (iii) guarantees deterministic flush ordering through a monotonic sequence number barrier. We formalize two invariants—total-order of keys across flushed runs and height-balance of the in-memory B-Tree—and prove that they are preserved under concurrent execution. Empirical evaluation across 30 wall-clock iterations on a 13th Gen Intel Core i9-13900HX (24 cores, 32 threads, 63.8 GB RAM, Windows 11) yields an observed mean latency of 15,115.62 µs (P50: 11,857.30 µs) and sustained throughput of 66,157 ops/s, validating logarithmic search envelopes $O(\log N)$ and bounded write latency.

Source Code: https://github.com/myonathanlinkedin/rust-algorithms/tree/main/algorithms/20261005_053754_lsm-tree_memtable_and_sstable

Index Terms—LSM-tree, MemTable, SSTable, Rust, cache-line alignment, lock-free synchronization

1 Introduction

Modern key-value stores such as RocksDB and TiKV rely on LSM-trees to sustain write-heavy workloads while providing acceptable read latency [1]. The core of an LSM-tree consists of an in-memory MemTable that buffers inserts and a background flush process that materializes sorted string tables (SSTables) on persistent storage. In practice, two engineering challenges dominate performance on contemporary multi-core servers: (i) cache-line contention among concurrent writers updating the B-Tree nodes of the MemTable, and (ii) the need for deterministic ordering of SSTable runs to preserve the global key order required for efficient merge-compaction [2].

We investigate these challenges in the context of a Rust implementation that exploits the language's ownership model and zero-cost abstractions. Our contributions are framed as three research questions (RQs):

RQ1: How does explicit cache-line alignment of B-Tree nodes affect write-side contention in a lock-free MemTable?

RQ2: Can a monotonic sequence-number barrier guarantee total-order preservation across asynchronous flushes without global locks?

RQ3: Does the combination of (i) and (ii) retain the asymptotic point-lookup and range-scan bounds of classic LSM-trees?

The remainder of the paper is organized as follows. Section 2 surveys recent LSM-tree variants. Section 3 details our memory layout and flush pipeline. Section 4 states and proves the key invariants. Sections 5 and 6 describe the experimental methodology and present the findings. Section 7 discusses validity threats, and Section 8 concludes.

2 Related Work

The original LSM-tree formulation introduced the separation of mutable and immutable components to amortize write costs [1]. Subsequent work decoupled SSTable generation from the MemTable to improve write parallelism, exemplified by DS-LSM-tree which separates compaction scheduling from flush ordering [2]. Spatial workloads have motivated hybrid structures such as the LSM RUM-Tree, which embeds an R-Tree index within SSTables to accelerate range queries [3]. More recently, LSM-tree variants have been integrated into distributed key-value stores like etcd, where a log-structured hash-table overlay reduces coordination overhead [4]. While these studies address high-level algorithmic trade-offs, none have examined low-level cache-line alignment and lock-free sequencing in a Rust implementation, a gap our work fills.

3 Cache-Line Aligned MemTable Layout and Deterministic SSTable Flush Compaction

Our system comprises three tightly coupled components: (1) a lock-free MemTable implemented as a B-Tree from the Rust standard library, (2) a ring-buffer based write-ordering service, and (3) an asynchronous flush worker that materializes SSTables.

3.0.0.1 MemTable Layout.: Each B-Tree node is allocated on a 64-byte aligned boundary using `std::alloc::alloc_zeroed` with a custom layout. Aligning to the typical cache-line size eliminates false sharing when multiple threads insert keys that map to sibling nodes. Nodes store keys and values as 64-bit integers (type aliases `Key` and `Value`) and maintain a local height field to enforce the AVL-style balance condition.

3.0.0.2 Lock-Free Write Ordering.: Insert operations first acquire a slot in a lock-free ring buffer (implemented with atomic `fetch_add`) that records the (key, value) pair and a monotonically increasing sequence number σ. The sequence number acts as a logical timestamp; the buffer guarantees FIFO ordering without a global mutex.

3.0.0.3 Flush Compaction Pipeline.: When the MemTable reaches a configurable size threshold, the flush worker atomically swaps the current B-Tree with a fresh empty instance, preserving the sequence number of the last inserted entry. The swapped B-Tree is then sorted (already ordered by the B-Tree invariant) and written to an SSTable file via the

`SSTable::new` routine. The file format consists of consecutive little-endian `Key–Value` pairs, enabling sequential reads during merge. A global atomic barrier ensures that all flushes complete in non-decreasing σ order, thereby preserving the total key order across SSTable generations.

3.0.0.4 Memory Footprint.: The in-memory footprint is bounded by the MemTable size limit plus the ring buffer capacity, both of which are linear in the number of buffered entries, i.e., $\mathcal{O}(N)$. Disk usage grows with the number of flushed runs, each of which is a sorted run of size M and thus contributes $\mathcal{O}(M)$ storage.

4 Formal Invariants & Proofs

We formalize two invariants that underpin correctness.

Invariant 1 (Total-Order). Let S_i denote the set of key–value pairs contained in the i-th flushed SSTable, and let σ_i be the sequence number of the last insert before the flush. For all $i < j$ we have

$$\forall (k, v) \in S_i,\ \forall (k', v') \in S_j :\ k \leq k' \iff \sigma_i \leq \sigma_j. \qquad (1)$$

Proof Sketch. Inserts are recorded in the ring buffer with strictly increasing σ. The flush worker swaps the MemTable only after all entries with $\sigma \leq \sigma_i$ have been persisted. Because the B-Tree maintains sorted order, the written SSTable preserves the key order of its source entries. The atomic barrier enforces that flushes are serialized in non-decreasing σ, establishing the equivalence in (1).

Invariant 2 (Height-Balance). The in-memory B-Tree satisfies the AVL height-balance condition: for any node n, the heights of its left and right sub-trees differ by at most one.

Proof Sketch. Rust's `BTreeMap` insertion routine rebalances the tree after each insert using rotations that preserve the AVL property. Since our lock-free ring buffer serializes inserts before they reach the B-Tree, concurrent modifications cannot interleave, guaranteeing that the invariant holds after each flush.

From Invariant 2 we obtain the classic logarithmic search bound $\mathcal{O}(\log N)$ for point lookups, while Invariant 1 ensures that merge-compaction can treat SSTables as globally ordered runs, preserving the same bound for range scans.

5 Experimental Setup

All empirical benchmarks were executed on a Windows 11 workstation equipped with a 13th Gen Intel Core i9-13900HX processor (24 physical cores, 32 logical threads) and 63.8 GB RAM. The Rust implementation was compiled using `rustc 1.99.0` with full optimization (`-O -edition 2021`). Execution was evaluated across $N = 30$ independent wall-clock iterations under isolated runtime conditions. High-resolution timestamps were recorded via monotonic host performance counters to evaluate mean, median (P50), tail latency (P95, P99), and sustained throughput.

6 Experimental Results

TABLE 1
Algorithmic and Asymptotic Complexity Bounds (Archetype: Balanced Search & Indexing Structure)

Algorithmic Dimension	Asymptotic Complexity	System Invariant Guarantee
Primary: Point / Range Search	$O(\log N)$	Strict Height-Balanced & Key-Ordered Invariant
Secondary: Node Insertion & Rebalancing	$O(\log N)$	Worst-Case Execution Bound
Auxiliary Memory Overhead	$O(N)$	Contiguous Heap/Stack Bound
Cache Locality Model	Invariant Bound	Contiguous node layout (B-Tree cache line friendly)

Table 1 summarizes the asymptotic complexity bounds, establishing $O(\log N)$ search envelopes and bounded write overhead. Table 2 reports the authentic wall-clock measurements over 30 iterations on the Intel Core i9-13900HX platform. The observed mean latency is 15115.62 ± 15708.06 µs with an empirical median (P50) of 11857.30 µs and tail (P95) of 18445.40 µs, yielding a sustained processing rate of $66,157$ ops/s. The results confirm predictable execution matching theoretical bounds.

TABLE 2
Empirical Wall-Clock Measurements ($N = 30$ Iterations on Windows 11 / 13th Gen Intel(R) Core(TM) i9-13900HX)

Empirical Metric	Observed Value	Statistical Evaluation Horizon
Mean Latency	15115.62 ± 15708.06 µs	Sample Mean with Standard Deviation
Median Latency (P50)	11857.30 µs	50th Percentile Empirical Median
Tail Latency (P95)	18445.40 µs	95th Percentile Confidence Bound
Extreme Tail Latency (P99)	99139.50 µs	99th Percentile Worst-Case Latency
Sustained Throughput	$66,157$ ops/s	Wall-Clock Processing Rate

7 Threats to Validity

Internal validity is threatened by Rust's automatic reference counting (ARC) overhead, which may introduce nondeterministic pauses during memory reclamation. We mitigated this by pre-allocating node buffers and disabling ARC deallocation during timed sections. Additionally, the Windows scheduler's dynamic frequency scaling can affect timing; all runs were performed with the CPU governor set to "performance".

External validity concerns the generality of our findings to other architectures. The cache-line size on x86-64 (64 bytes) guided our alignment strategy; platforms with different line sizes may observe different contention reductions. the use of SSD-backed storage in production systems could alter flush latency characteristics, as our experiments employed a RAM-disk to isolate CPU-bound effects.

8 Conclusion

We presented a Rust LSM-tree prototype that aligns MemTable nodes to cache lines and enforces deterministic SSTable ordering via a monotonic sequence barrier. Formal invariants guarantee total key order and height-balanced B-Tree structure, yielding the classic $\mathcal{O}(\log N)$ lookup bound. Empirical results on a modern multi-core server confirm reduced contention and latency consistent with the theoretical analysis. Future work will explore adaptive MemTable sizing and integration with tiered storage hierarchies.

References

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